THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $x = 4$ is a root of $x^2 - 9x + k = 0$, then $k =$ 1

(a) 22 **(b)** 20 **(c)** 21 **(d)** 19

2. The decimal expansion of $\frac{77}{210}$ is: 1

(a) terminating **(b)** non-terminating repeating

(c) non-terminating non-repeating **(d)** an integer

3. The remainder when $x^3 + 2x^2 - 3x + 4$ is divided by $x - 2$ is: 1

(a) 15 **(b)** 14 **(c)** -14 **(d)** 10

4. If $3x + 2y = 25$ and $2x + 3y = 20$, then $x + y =$ 1

(a) 11 **(b)** 8 **(c)** 10 **(d)** 9

5. The equation $x^2 + 6x + 8 = 0$ has: 1

(a) one real root and one imaginary root **(b)** no real roots

(c) two distinct real roots **(d)** two equal real roots

6. If $\sqrt{3}$ is irrational, then $\sqrt{3} \times \sqrt{3}$ is: 1

(a) irrational **(b)** non-terminating non-repeating

(c) rational **(d)** not defined

7. If the sum of the zeroes of $x^2 - kx + 12$ is 7, then the zeroes could be 3 and: 1

(a) 3 **(b)** 4 **(c)** 12 **(d)** 5

8. If $x + y = 9$ and $x - y = 3$, then $x^2 - y^2 =$ 1

(a) 29 **(b)** 27 **(c)** 26 **(d)** 28

9. The larger root of $x^2 - 11x + 24 = 0$ is: 1

(a) 7 **(b)** 10 **(c)** 8 **(d)** 9

10. The HCF of 45, 75 and 90 is: 1

(a) 15 **(b)** 14 **(c)** 17 **(d)** 16

11. If α, β are zeroes of $x^2 - 7x + 12$, then $\alpha^2 + \beta^2$ is: 1

(a) 25 **(b)** 24 **(c)** 49 **(d)** 26

12. The value of x in $5x + 3y = 29$, $4x + 7y = 37$ is: 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

13. The sum of the roots of $3x^2 + 7x - 6 = 0$ is: 1

(a) $-\frac{10}{3}$ **(b)** $-\frac{1}{3}$ **(c)** $-\frac{4}{3}$ **(d)** $-\frac{7}{3}$

14. The LCM of 10, 15 and 20 is: 1

(a) 300 **(b)** 61 **(c)** 60 **(d)** 30

15. If α, β are zeroes of $x^2 - 5x + 4$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is: 1

(a) $\frac{4}{5}$ **(b)** $-\frac{5}{4}$ **(c)** $\frac{9}{4}$ **(d)** $\frac{5}{4}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. If $x = 3$ is a zero of $x^2 - 8x + k$, find k. 2

17. For what value of k does the pair $kx + 2y = 5$, $6x + 1y = 3$ have a unique solution? 2

18. If $x = 4$ is a root of $x^2 - 9x + k = 0$, find k and the other root. 2

19. Without actual division, find how many decimal places $\frac{23}{200}$ has. 2

20. Find a quadratic polynomial whose zeroes are 2α and 2β, where α, β are zeroes of $x^2 - 3x + 2$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the nature of the roots of $3x^2 - 7x + 2 = 0$ and, if real, find them. 3

22. Show that $(5 + 2\sqrt{3})(5 - 2\sqrt{3})$ is rational but $5 + 2\sqrt{3}$ is irrational. 3

23. If 6 is a zero of the polynomial $x^2 + kx + 30$, find k and the other zero. 3

24. Find the value of k for which the pair $kx + 1y = k - 1$ and $4x + ky = k$ has infinitely many solutions. 3

OR

Solve by completing the square: $x^2 - 10x + 21 = 0$. 3

25. The HCF of two numbers is 12 and their LCM is 504. If one number is 84, find the other. 3

OR

If α, β are zeroes of $x^2 - 6x + 5$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ and $\alpha^2 + \beta^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A rectangular courtyard is 24 m by 16 m. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

OR

Find k if the product of the zeroes of $x^2 - 5x + k$ equals 2 times their sum, and find the polynomial. 5

27. A fraction becomes $\frac{1}{2}$ when 1 is added to both the numerator and the denominator, and becomes $\frac{5}{9}$ when 2 is added to both. Find the fraction. 5

OR

The altitude of a right triangle is 1 cm less than its base and the hypotenuse is 29 cm. Find the base and altitude. 5

END OF QUESTION PAPER **Total: 50 Marks**

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